

Joint Detection of AI-Generated Images and Post-Processing Alterations

Master of Science in Artificial Intelligence

Students: Paola Cortés Domínguez & Carmen Rojas Gallego

Professor: Irene Amerini | Sapienza University of Rome

PRESENTATION OUTLINE

- ✓ **Problem Statement:** Authenticity in real scenarios.
- ✓ **State of the Art:** Existing forensic methods.
- ✓ **Proposed Method:** Multi-Task Learning (MTL).
- ✓ **Dataset:** Sub-selection and balancing.
- ✓ **Experimental Setup:** Parameters and Loss.
- ✓ **Model Evaluation:** Quantitative Results (88%).
- ✓ **Conclusions:** Final thoughts and future work.
- ✓ **References:** Sources of inspiration.

PROBLEM STATEMENT

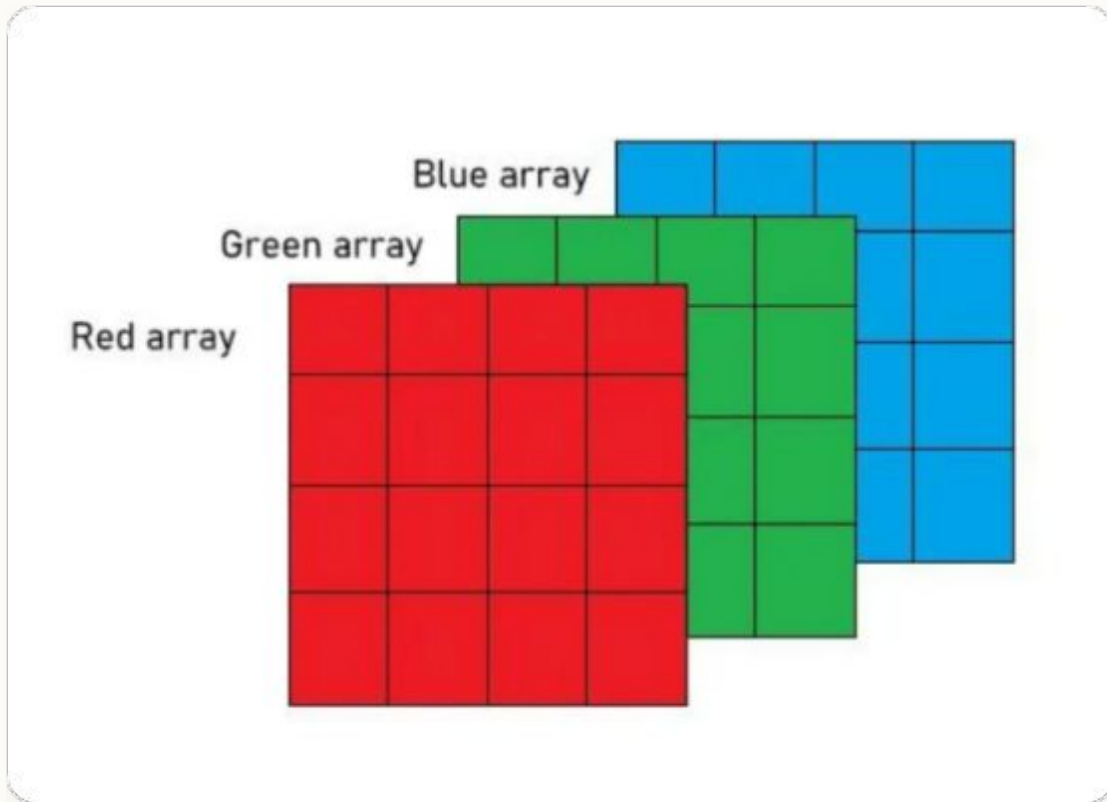
In modern digital platforms, AI-generated images are seamlessly mixed with real photographs.

The Challenge: Identifying authenticity while accounting for common transformations:

- ✓ Internet Transfer (Compression artifacts).
- ✓ Screen Re-digitization (Moiré patterns).
- ✓ Original high-quality acquisition.



STATE OF THE ART



- ✓ **Artifact Detection:** Searching for upsampling noise or checkerboard patterns in GAN images.
- ✓ **Frequency Domain:** Analyzing the spectrum to find anomalies in high frequencies.
- ✓ **Robustness Issue:** Most detectors fail when images undergo common social media processing (resizing, compression).

PROPOSED METHOD: MULTI-TASK LEARNING

Instead of using independent networks, we propose a **unified MTL framework** using a **ResNet18 backbone**.

The network predicts two labels simultaneously:

- ✓ **Class:** Real vs AI-Generated (Fake).
- ✓ **Scenario:** Original, Transfer, or Screen-Re-digitized.



DATASET: RRDATASET SUB-SELECTION



1,800

Images

- ✓ **Balanced Selection:** 300 images per category.
- ✓ **Efficient Training:** Balanced data prevents the model from collapsing into majority classes.
- ✓ **Data Split:** 80% (1,440 images) for Training and 20% (360 images) for Validation.
- ✓ **Pre-processing:** 224x224 resize and ImageNet normalization.

EXPERIMENTAL SETUP

Configuration	Setting / Value
Model Backbone	ResNet18 (Pre-trained on ImageNet)
Optimizer	Adam (Learning Rate: 10^{-4})
Batch Size	32 Images
Epochs	10 Epochs
Hardware	Google Colab GPU

JOINT LOSS OPTIMIZATION

The model is optimized using a weighted sum of two independent Cross-Entropy losses:

$$L_{\text{total}} = 0.5L_{\text{class}} + 0.5L_{\text{trans}}$$

Both tasks contribute equally to the gradient updates, ensuring a robust shared feature representation in the ResNet backbone.

MODEL EVALUATION: RESULTS

Overall Accuracy

88%

Precision (AI Fake)

90%

Recall (Real)

89%

* Metrics evaluated on the independent 360-image validation set.

Class	Precision	Recall	F1
Real	0.85	0.89	0.87
AI (Fake)	0.90	0.87	0.88
Accuracy	0.89		

CONFUSION MATRIX ANALYSIS

Detailed Breakdown:

- ✓ **Real Images:** 147 correctly classified out of 165.
- ✓ **AI Fake Images:** 169 correctly detected out of 195.
- ✓ **High Precision:** 90% Precision for AI images suggests very few false alarms.
- ✓ **Balanced Error:** Misclassifications are distributed across both tasks.



Validation Size: 360 Samples

The MTL framework effectively learns forensic features even under internet transmission conditions.

CONCLUSIONS & FUTURE WORK

Key Findings

The Multi-Task Learning framework constitutes a viable approach for image forensics, achieving **88% accuracy** while sharing a unified feature representation.

Limitations:

- ✓ Hardware constraints limited the subset size.
- ✓ Fixed loss-weighting strategy.

Future Work:

- ✓ Alternative backbones (EfficientNet, Transformers).
- ✓ Ablation study on loss weights (λ).

REFERENCES

- ✓ [1] **RRDataset**: Robust Real-world Image Forensics Dataset.
- ✓ [2] **He et al. (2016)**: Deep Residual Learning for Image Recognition (ResNet).
- ✓ [3] **Scikit-Learn Documentation**: Classification reports and metrics calculation.
- ✓ [4] **Pytorch Documentation**: Transfer Learning and Multi-Task optimization strategies.

Thank You!

Questions?